

Natural-Language versus Symbolic Messaging for Ground-Aerial Open-Vocabulary Search in a Toy World

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<https://github.com/mihir-chauhan/nl-ground-aerial-ovs-search>

Abstract—This document was produced by the platform’s demonstration backend, which uses no language model. It runs one fixed experiment, a comparison of gradient descent, heavy-ball momentum [1] and Adam [2] on synthetic ill-conditioned least squares, to exercise the full pipeline: repository, runs, tables, figures, citation checks and this paper. It makes no research claim and does not address the seed it was started from.

Index Terms—pipeline demonstration, gradient descent, momentum

I. Introduction

First-order optimisers are the standard tools for fitting models [3]. This demonstration compares three of them on a problem whose conditioning is controlled, so that every stage of the platform has real inputs.

II. Related Work

Heavy-ball momentum was introduced by Polyak [1]. Adam [2] adapts per-coordinate step sizes. Ruder [3] surveys these methods.

III. Method

We minimise $f(w) = \frac{1}{2n} \|Xw - y\|^2$. Gradient descent and momentum use step $1/L$, with L the largest eigenvalue of $X^\top X/n$; momentum adds a velocity term with coefficient β . Adam uses its default moments.

IV. Experimental Setup

Synthetic data with $n = 200$ and $d = 20$; column scales set the condition number to 10 or 1000. Each system runs 400 steps on three seeds. We report the excess loss after the last step and the number of steps to reach a fixed tolerance. All baselines are reimplemented in numpy.

Produced by the Stemma autonomous AI research pipeline from a research seed and direction given by the named author: the pipeline ran the experiments and wrote the text. Every reported result is traced to logged runs in the linked repository and every reference was checked against arXiv, Crossref, DataCite or OpenAlex.

TABLE I

Steps to tolerance and final excess loss (mean $\pm$ std over seeds).

Method	Task	steps_to_tol $\downarrow$	final_loss $\downarrow$	n
GD	cond10	42.000 ± 3.606	-0.000 ± 0.000	3
Adam	cond10	100.667 ± 2.082	0.000 ± 0.000	3
Momentum	cond10	90.333 ± 2.082	0.000 ± 0.000	3
GD	cond1000	400.000 ± 0.000	1.112 ± 0.762	3
Adam	cond1000	136.667 ± 16.258	0.000 ± 0.000	3
Momentum	cond1000	300.000 ± 39.000	0.000 ± 0.000	3

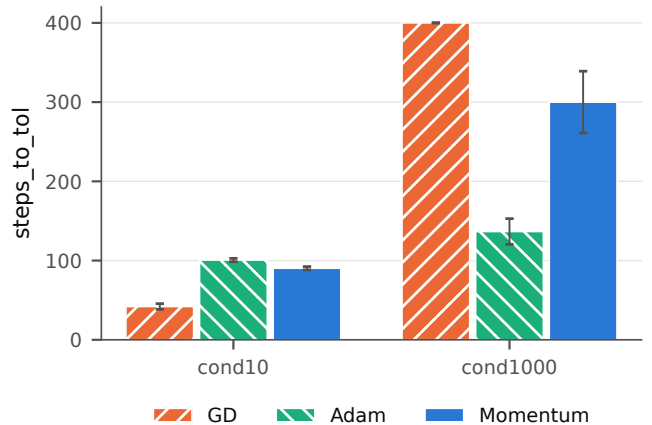


Fig. 1. Steps to tolerance per task.

V. Results

Table I and Fig. 1 report all systems on both tasks. The steps-to-tolerance metric is the step at which a run first reached the tolerance, or the step budget when it never did, so a value equal to the budget means the tolerance was not reached. On the harder task its mean is 300.0 for momentum, 400.0 for gradient descent and 136.7 for Adam (recorded values, not typed).

VI. Ablations and Analysis

Fig. 2 shows the effect of the momentum coefficient on the harder task.

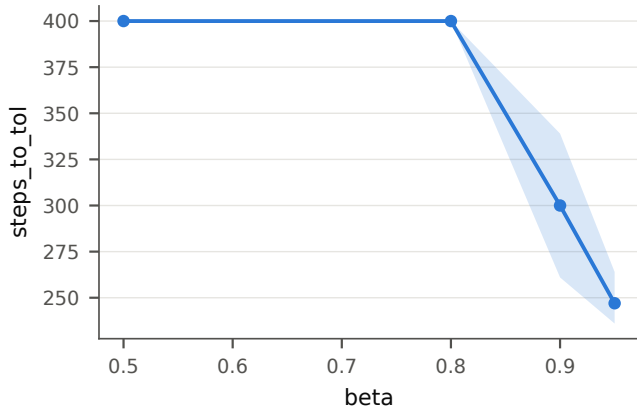


Fig. 2. Momentum coefficient sweep.

VII. Limitations

This is a pipeline demonstration on a toy problem with three seeds. It supports no conclusion beyond the table above.

VIII. Conclusion

The pipeline ran end to end: runs were logged, tables and figures generated, and references verified.

References

- [1] B. T. Polyak, “Some methods of speeding up the convergence of iteration methods,” USSR Computational Mathematics and Mathematical Physics, 1964. [Online]. Available: [https://doi.org/10.1016/0041-5553\(64\)90137-5](https://doi.org/10.1016/0041-5553(64)90137-5)
- [2] D. P. Kingma and J. Ba, “Adam: A Method for Stochastic Optimization,” arXiv preprint arXiv:1412.6980, 2014. [Online]. Available: <https://arxiv.org/abs/1412.6980>
- [3] S. Ruder, “An overview of gradient descent optimization algorithms,” arXiv preprint arXiv:1609.04747, 2016. [Online]. Available: <https://arxiv.org/abs/1609.04747>